

Synthetic Image Generation

https://github.com/aiyshwary/Synthetic_waste_image_generation

Python

Computer Vision

OpenCV

Data Augmentation

OVERVIEW

An automated pipeline for generating high-quality synthetic training images for object detection, addressing class-imbalanced datasets by compositing annotated object crops onto varied backgrounds with randomized augmentations.

IMPACT

Built an end-to-end synthetic image generation pipeline that composites real object crops onto diverse backgrounds with randomized augmentations, producing annotated training data that reduces dependency on costly manual data collection.

KEY DETAILS

- Extracts foreground object crops from existing annotated datasets using bounding box and mask information.

- Composites crops onto varied background images with randomized scale, rotation, and position transformations.

- Automatically generates YOLO/Pascal VOC-compatible annotation files alongside each synthetic image.

- Targets underrepresented classes to balance dataset distributions for improved model generalization.

- Validated improvements in downstream detection model performance on augmented versus original-only training sets.